

CYBERSECURITY TESTING LAB SETUP

Task WK1-PM1

Lab Setup for Cyber Security & Ethical Hacking Practice

Name	Daniyal Ahmed
Course	Cybersecurity & Ethical Hacking
Task ID	WK1-PM1
Intern ID	NW-83-4UP
Virtualization Tool Used	VMware Workstation Pro 17
Attacking Machine	Kali Linux (VMware image)

1. Objective

The objective of this task was to build a cybersecurity testing lab on a personal laptop/PC, using a hypervisor to host an attacking machine (Kali Linux) on an isolated, NAT-based internal network (10.0.0.0/24), with Kali configured at a static IP of 10.0.0.2/24, full internet access, and shared clipboard / shared-folder access with the host.

The reference lab guide (WK1-PM1) illustrates the setup using Oracle VirtualBox. For this submission, VMware Workstation Pro 17 was used instead, as it provides the same core capabilities (NAT networking, adapter configuration, guest isolation, shared folders, and clipboard integration) through an equivalent set of settings. The steps below map each VirtualBox concept from the reference guide to its VMware Workstation equivalent, and are illustrated with screenshots taken while completing the task.

2. Tools & Environment

- Hypervisor: VMware Workstation Pro 17
- Attacking machine: Kali Linux 2026.2 (official pre-built VMware image)
- Network mode: NAT (VMware's built-in NAT service, equivalent to VirtualBox's NATNetwork)
- Target subnet: 10.0.0.0/24
- Kali Linux static IP: 10.0.0.2/24, Gateway 10.0.0.1, DNS 8.8.8.8

3. Setup Steps & Screenshots

Step 1: Install VMware Workstation Pro

VMware Workstation Pro was downloaded and installed as the hypervisor/base platform for the lab, replacing VirtualBox used in the reference guide. This is the equivalent of Step 2 in the reference PDF ("Download & install Virtualbox").



Figure 1 — VMware Workstation download page, showing the recommended VMware build available for download.

Step 2: Launch VMware Workstation Pro

After installation, VMware Workstation Pro 17 was launched. The Workstation home screen provides options to create a new VM, open an existing VM, or connect to a remote server — this is the VMware equivalent of the VirtualBox Manager main window.

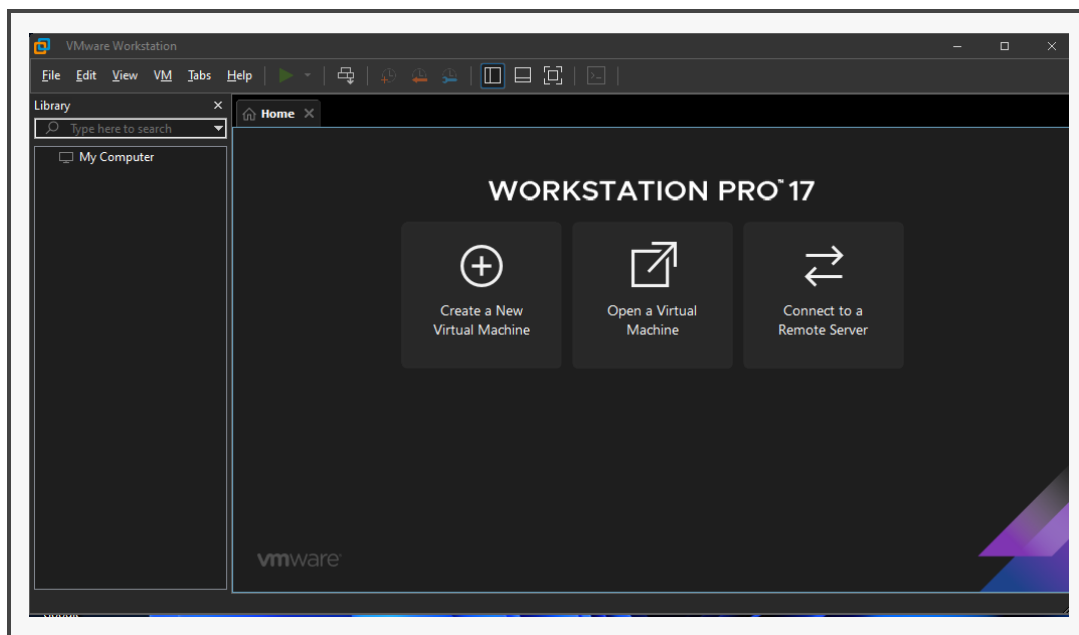


Figure 2 — VMware Workstation Pro 17 home screen after installation.

Step 3: Download the Kali Linux Pre-built VM (VMware image)

Instead of importing a VirtualBox OVA (Step 4 in the reference guide), the pre-built Kali Linux image built specifically for VMware was downloaded from the official Kali "Pre-built Virtual Machines" page (kali.org/get-kali). This image comes ready to run in VMware without a manual OS installation, using the default credentials kali/kali.

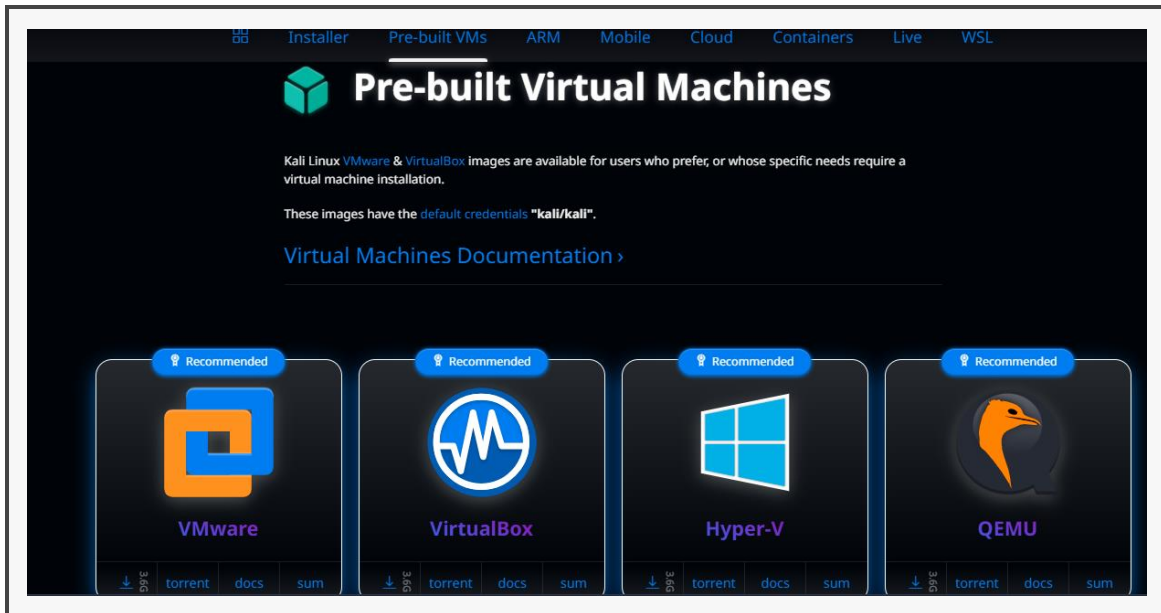


Figure 3 — Kali Linux official download page, VMware image selected from the list of pre-built virtual machines.

Step 4: Import / Open the Kali Linux VM in VMware

The downloaded Kali Linux VMware package was opened directly from VMware Workstation using File > Open, selecting the extracted .vmx configuration file (kali-linux-2026.2-vmware-amd64). This registers the VM in the Workstation library, equivalent to the "Import Appliance" step in VirtualBox.

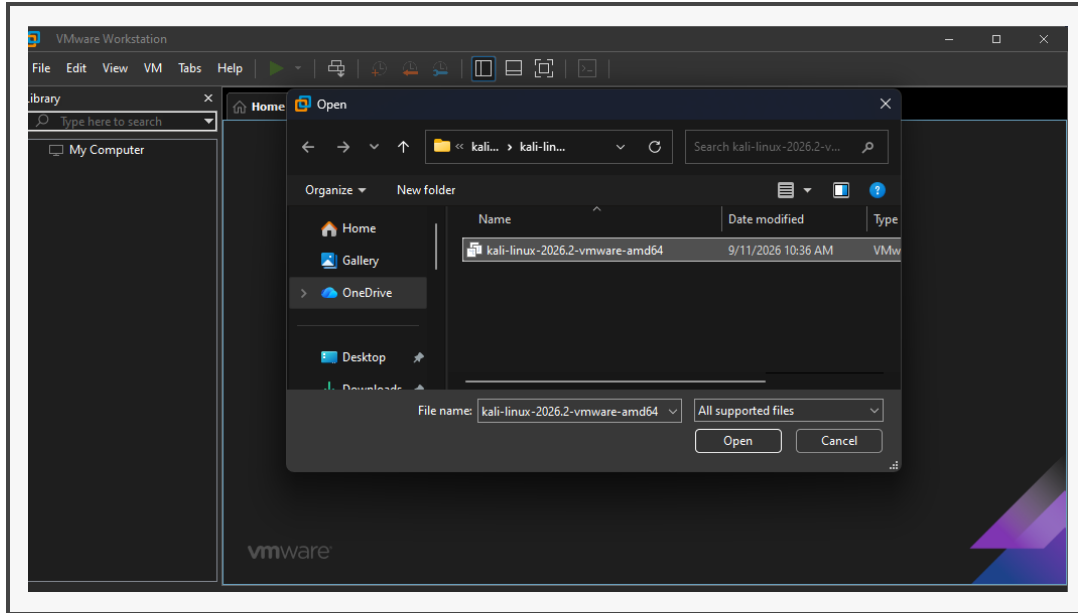


Figure 4 — Opening the Kali Linux VM package in VMware Workstation via the Open dialog.

Step 5: Configure the Network Adapter to NAT

In the Kali Linux VM's settings, the Network Adapter was set to NAT ("used to share the host's IP address"), with "Connect at power on" enabled. This is the VMware equivalent of attaching the VirtualBox NATNetwork adapter and places the VM on an isolated internal subnet (VMware's default NAT network, VMnet8) that is translated to the host's internet connection — matching the reference guide's requirement to use NATNetwork in 10.0.0.0/24.

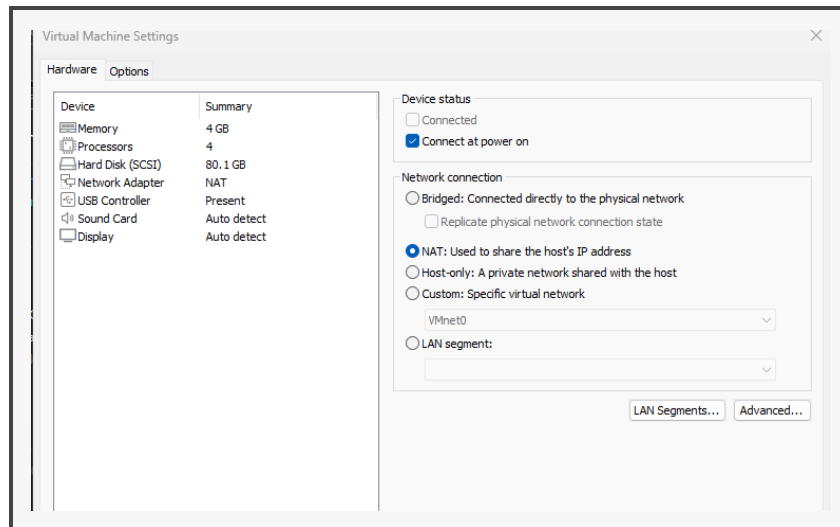


Figure 5 — Kali Linux VM settings: Network Adapter set to NAT, connected at power on.

Step 6: Configure Display / Graphics Settings

The VM's Display settings were reviewed and 3D graphics acceleration was enabled, with the display set to use the host's monitor settings and 256 MB of graphics memory allocated. This ensures the Kali desktop environment runs smoothly inside the VMware window.

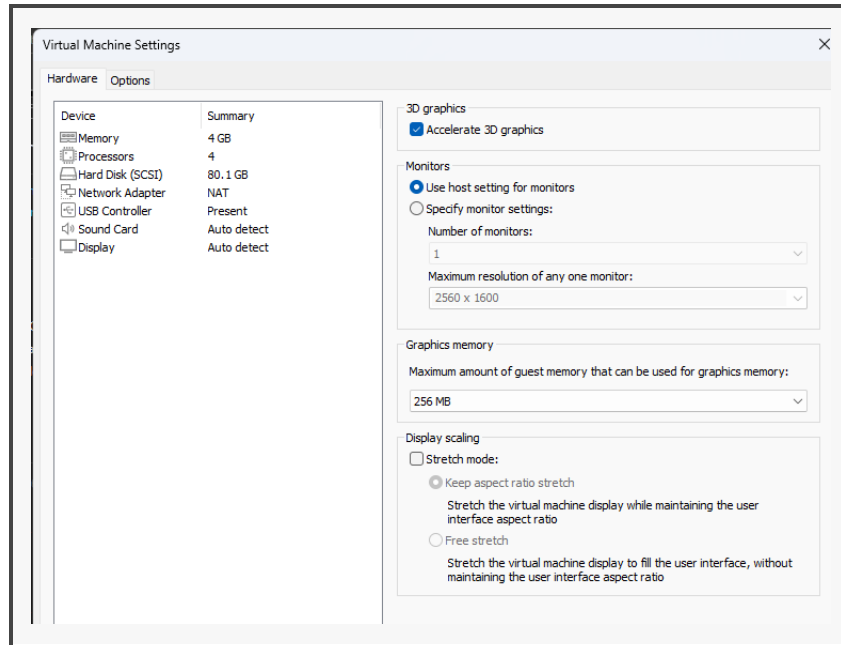


Figure 6 — Kali Linux VM settings: Display configuration with 3D acceleration enabled.

Step 7: Assign the Static IP Address to Kali Linux

Inside the running Kali Linux VM, the network connection's IPv4 settings were edited (Network Manager > Editing Wired connection 1 > IPv4 Settings) to add the additional static address 10.0.0.2/24 with gateway 10.0.0.1 and DNS server 8.8.8.8, in addition to the default DHCP method. This satisfies the requirement that Kali Linux should be reachable at 10.0.0.2/24 while still allowing full internet access through the NAT gateway.

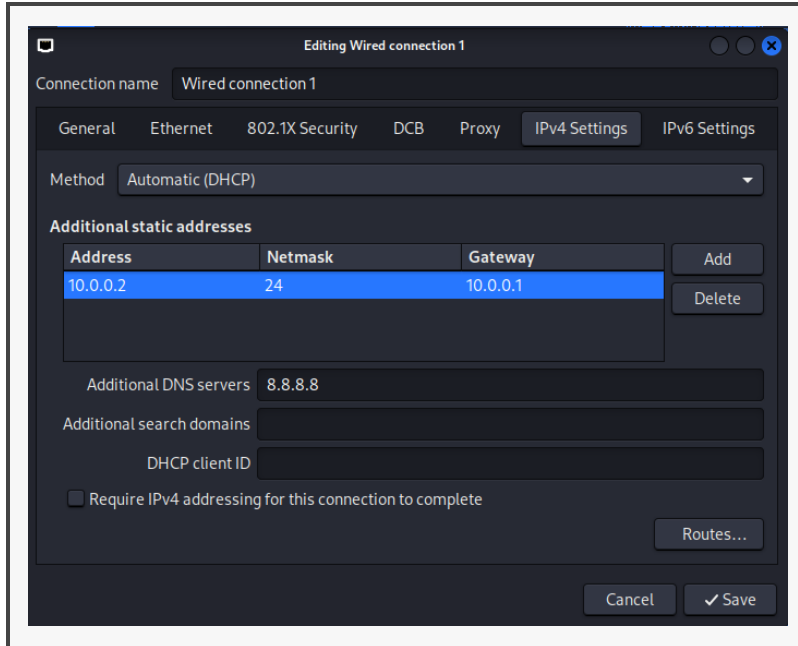


Figure 7 — Kali Linux Network Manager, IPv4 Settings tab showing the static address 10.0.0.2/24, gateway 10.0.0.1, and DNS 8.8.8.8.

4. Task Requirements Checklist

- VMware Workstation Pro installed and used as the hypervisor — Done
- Kali Linux imported as the attacking/hacker machine — Done
- Network configured in subnet 10.0.0.0/24 using NAT — Done
- Kali Linux assigned static IP 10.0.0.2/24 — Done
- Kali Linux configured with full internet access via NAT gateway (10.0.0.1) — Done
- Clipboard, drag-and-drop, and shared folders — to be enabled via VM Settings > Options > Shared Folders / Guest Isolation (VMware Tools required)

5. Notes

All configuration steps mirror the reference lab guide (WK1-PM1), substituting VMware Workstation Pro 17 for Oracle VirtualBox. Functionally equivalent settings were used throughout: VMware's NAT network in place of VirtualBox's NATNetwork, and VMware's per-VM Network Adapter / Display / Shared Folders panels in place of VirtualBox's Network / Shared Folders tabs. A snapshot of the completed VM should be taken (VM > Snapshot > Take Snapshot) to preserve this working state before proceeding to Phase 2 of the lab.